CSE 4137

Machine Learning
Module-01
Regression



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Topics: Supervised ML

Week-03

- Regression Metrics
- Linear Regression with Metrics
- Multivariate Linear Regression

Metrics: Supervised ML

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Regression Metrics

Regression metrics are quantitative measures used to evaluate the nice of a regression model. Scikit-analyze provides several metrics, each with its own strengths and boundaries, to assess how well a model suits the statistics.

Types of Regression Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R-squared (R^2) Score
- Root Mean Squared Error (RMSE)

Mean Absolute Error (MAE)

In the fields of statistics and machine learning, the Mean Absolute Error (MAE) is a frequently employed metric. It's a measurement of the typical absolute discrepancies between a dataset's actual values and projected values.

The formula to calculate MAE for a data with "n" data points is:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|x_i - y_i|$$

Where:

- x_i represents the actual or observed values for the i-th data point.
- y_i represents the predicted value for the i-th data point.

Example

```
from sklearn.metrics import mean_absolute_error

true_values = [2.5, 3.7, 1.8, 4.0, 5.2]
predicted_values = [2.1, 3.9, 1.7, 3.8, 5.0]

mae = mean_absolute_error(true_values, predicted_values)
print("Mean Absolute Error:", mae)
```

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Mean Squared Error (MSE)

A popular metric in statistics and machine learning is the Mean Squared Error (MSE). It measures the square root of the average discrepancies between a dataset's actual values and projected values. MSE is frequently utilized in regression issues and is used to assess how well predictive models work.

For a dataset containing 'n' data points, the MSE calculation formula is:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2$$

where:

xi represents the actual or observed value for the i-th data point.

yi represents the predicted value for the i-th data point.

Example

```
from sklearn.metrics import mean_squared_error

true_values = [2.5, 3.7, 1.8, 4.0, 5.2]
predicted_values = [2.1, 3.9, 1.7, 3.8, 5.0]

mse = mean_squared_error(true_values, predicted_values)
print("Mean Squared Error:", mse)
```

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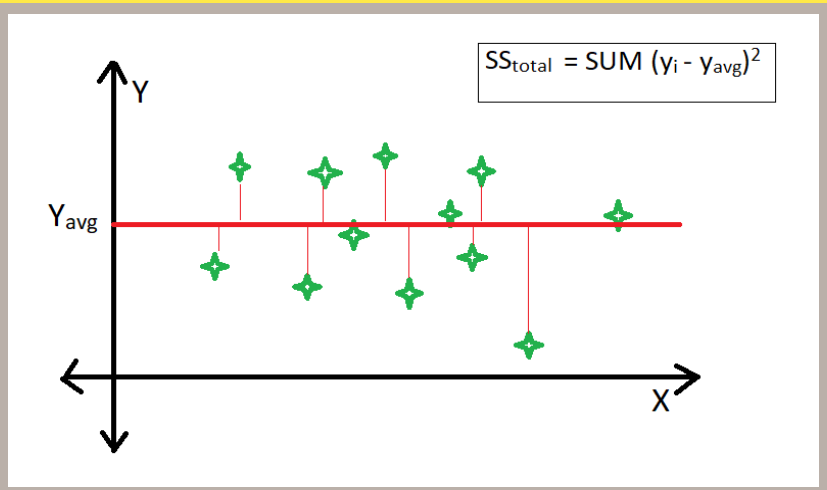
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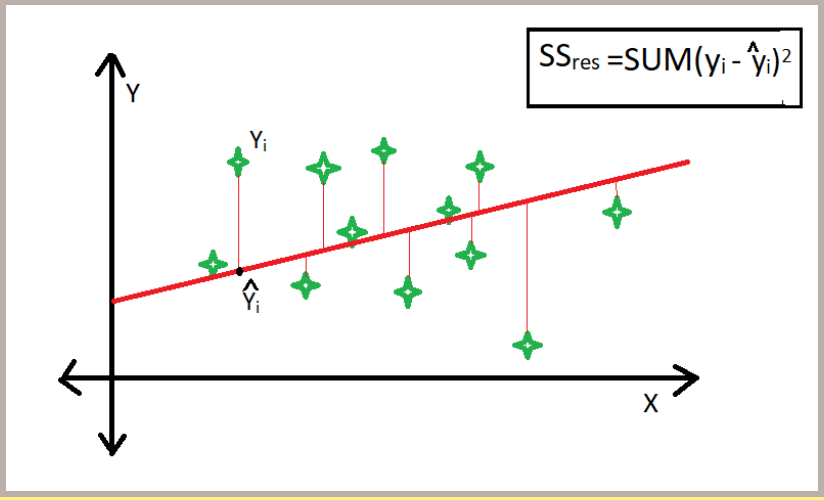
R-Squared Score (R2 Score)

A statistical metric frequently used to assess the goodness of fit of a regression model is the R-squared (R2) score, also referred to as the coefficient of determination. It quantifies the percentage of the dependent variable's variation that the model's independent variables contribute to. R2 is a useful statistic for evaluating the overall effectiveness and explanatory power of a regression model.

R-square is a comparison of the residual sum of squares *(SS_{res})* with the total sum of squares*(SS_{tot})*. The total sum of squares is calculated by summation of squares of perpendicular distance between data points and the average line.

The residual sum of squares is calculated by the summation of squares of perpendicular distance between data points and the best-fitted line.





R square is calculated by using the formula :

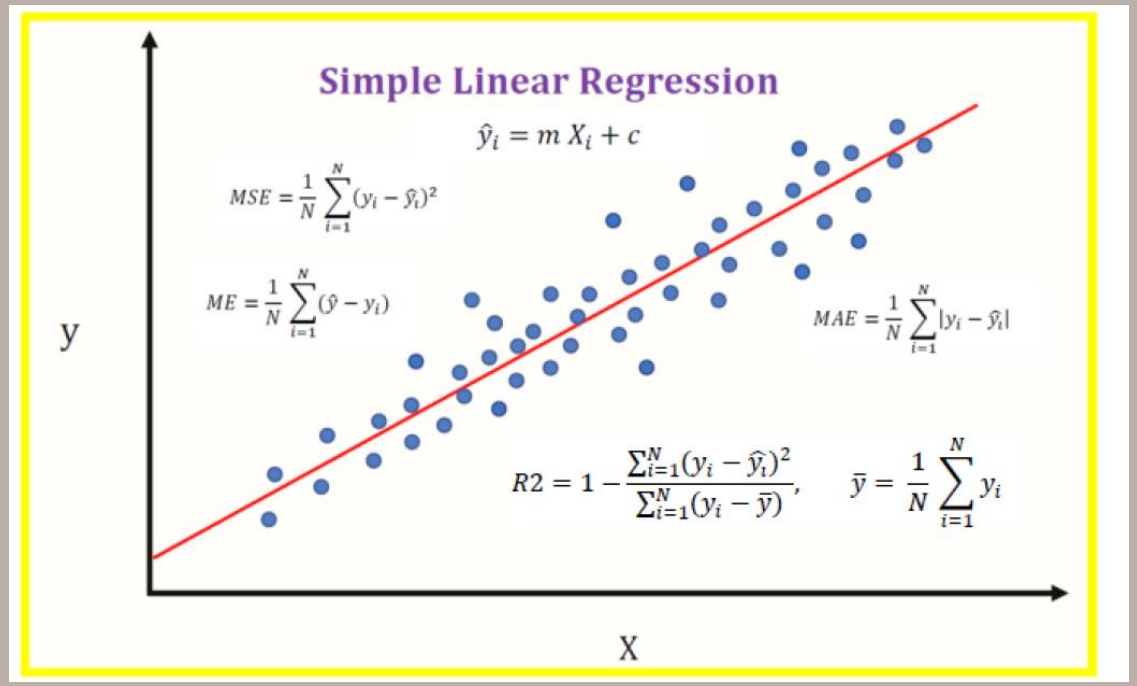
$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

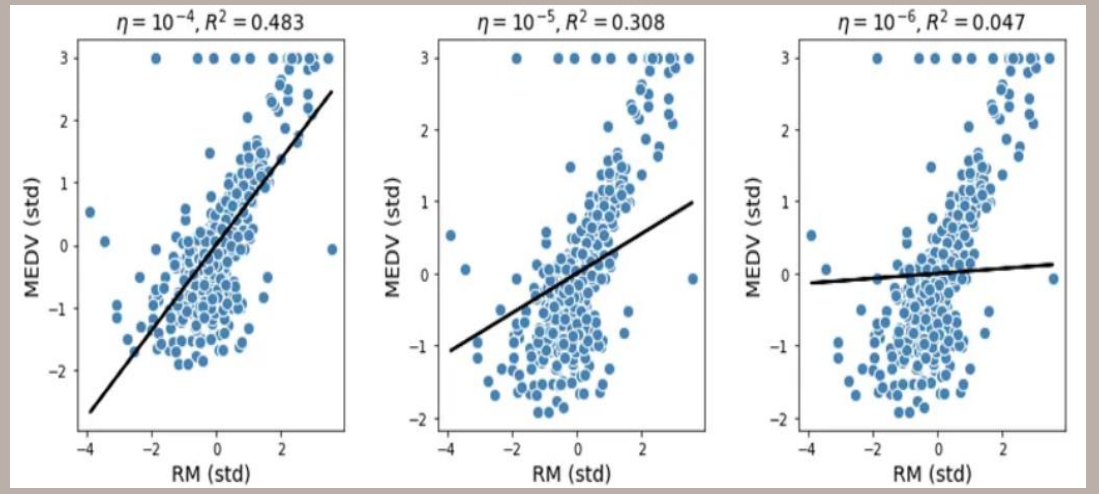
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R2 Score Contd

Where SS_{res} is the residual sum of squares and SS_{tot} is the total sum of squares.
The goodness of fit of regression models can be analyzed on the basis of the R-square method. The more the value of the r-square near 1, the better the model is.





Look at regression line and R2_Score

Example

```
from sklearn.metrics import r2_score

true_values = [2.5, 3.7, 1.8, 4.0, 5.2]
predicted_values = [2.1, 3.9, 1.7, 3.8, 5.0]

r2 = r2_score(true_values, predicted_values)
print("R-squared (R²) Score:", r2)
```

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Root Mean Squared Error

RMSE stands for Root Mean Squared Error. It is a usually used metric in regression analysis and machine learning to measure the accuracy or goodness of fit of a predictive model, especially when the predictions are continuous numerical values.

The formula for RMSE for a data with 'n' data points is as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2}$$

Where:

RMSE is the Root Mean Squared Error.
xi represents the actual or observed value for the i-th data point.
yi represents the predicted value for the i-th data point.

Example

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
import numpy as np

# Sample data
true_prices = np.array([250000, 300000, 200000, 400000, 350000])
predicted_prices = np.array([240000, 310000, 210000, 380000, 340000])

# Calculate RMSE
rmse = np.sqrt(mean_squared_error(true_prices, predicted_prices))

print("Root Mean Squared Error (RMSE):", rmse)
```

Example: Regression with Metrics

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Example

Use the dataset ‘placement.csv’ to find the package using cgpa.

Initial imports

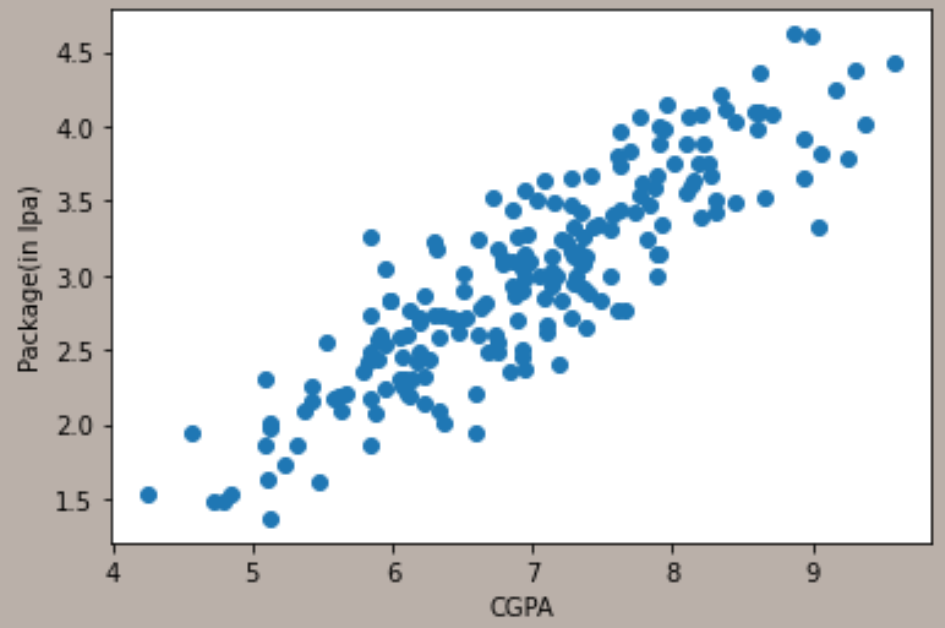
```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

1. Load Dataset

```
df = pd.read_csv('F:\\python\\placement.csv')
df
df.shape
```

2. Plot Data Points

```
plt.scatter(df['cgpa'],df['package'])
plt.xlabel('CGPA')
plt.ylabel('Package(in lpa)')
```



3. Split Dataset into Train and Test

```
X = df.iloc[:,0:1]
y = df.iloc[:,-1]
```

```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test =
train_test_split(X,y,test_size=0.2,random_state=2)
```

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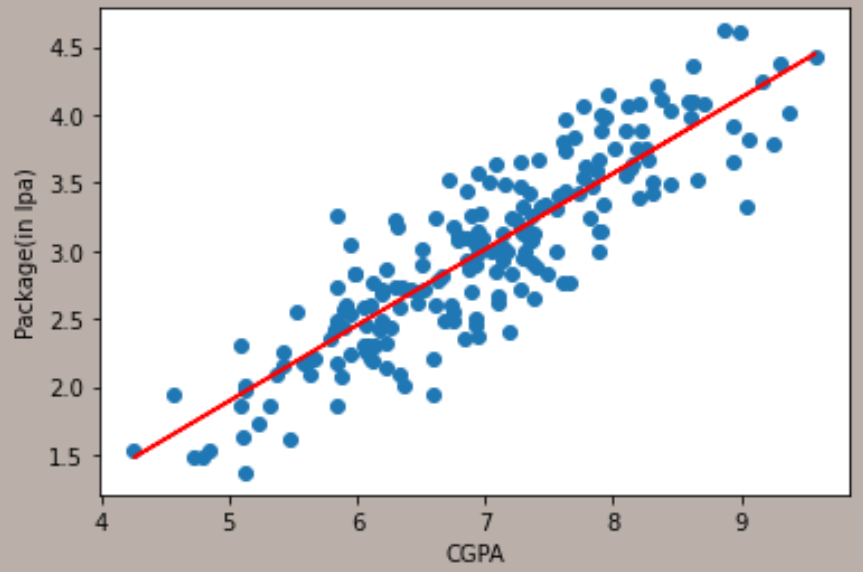
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4. Train the Model

```
from sklearn import  linear_model
lr = linear_model.LinearRegression()
lr.fit(X_train,y_train)
```

5. Plot the Regression Line

```
plt.scatter(df['cgpa'],df['package'])
plt.plot(X_train,lr.predict(X_train),color='red')
plt.xlabel('CGPA')
plt.ylabel('Package(in lpa)')
```



6. Test the Model

```
y_pred = lr.predict(X_test)
```

Test the Model

```
lr.predict([[7.5]])
```

7. Find Metrics

```
from sklearn.metrics import
mean_absolute_error,mean_squared_error,r2_score
print("MAE",mean_absolute_error(y_test,y_pred))
print("MSE",mean_squared_error(y_test,y_pred))
print("RMSE",np.sqrt(mean_squared_error(y_test,y
_pred)))
print(R2-Score",r2_score(y_test,y_pred))
```

8. Implement the Model

```
p=int(input("Enter cgpa:"))
q=[[p]]
lr.predict(q)
```

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End of Chapter